



REVA
UNIVERSITY
Bengaluru, India



School of Computer Science and Engineering

MINI PROJECT REPORT ON

“SANKETA – Real-Time Sign Language Recognition and Learning System”

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In partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

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Under the guidance of Professor

Dr. Muthireddy P

School of Computer Science & Engineering

Academic Year: 2025 -26



REVA
UNIVERSITY
Bengaluru, India

School of Computer Science and Engineering

CERTIFICATE

This is to certify that the mini project (B22CI0605) entitled “**SANKETA – Real-Time Sign Language Recognition and Learning System**” is a Bonafide work carried out by

Pujari Varshitha (R23EJ094)

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in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Information Technology of REVA University, Bengaluru during the academic year 2025-26

This work has been carried out under my supervision and guidance. To the best of my knowledge, this work has not been submitted previously for the award of any degree or diploma.

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Place: Bangalore Date:

DECLARATION

We hereby declare that the mini project entitled “SANKETA – Real-Time Sign Language Recognition and Learning System” is a Bonafide work carried out by us in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Information Technology of REVA University, Bengaluru during the academic year 2025-26.

We further declare that this work has not been submitted previously for the award of any degree or diploma in any university or institution. All sources of information used in this project have been properly acknowledged.

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ABSTRACT

Communication plays a vital role in connecting people and enabling meaningful interaction in daily life. However, communication barriers still exist between hearing individuals and the Deaf and Hard-of-Hearing community due to limited understanding of sign language. Traditional sign language learning and recognition systems often require specialized hardware, expensive infrastructure, or provide limited interactive learning support. Therefore, there is a need for an accessible, intelligent, and real-time system that can support both sign language recognition and learning.

The proposed project, “SANKETA – Real-Time Sign Language Recognition and Learning System,” is an AI-powered browser-based platform designed to bridge the communication gap between hearing individuals and the Deaf and Hard-of-Hearing community. The system integrates real-time hand gesture recognition with an interactive learning environment to provide both communication assistance and educational support. The system uses Media Pipe Hands for extracting 21 three-dimensional hand landmarks from webcam input. These landmarks are processed using feature engineering techniques such as inter-point distance calculation, joint angle analysis, and relative positioning. A lightweight k-Nearest Neighbour (KNN) machine learning algorithm is used for gesture classification. The system also includes temporal gesture processing using a sliding window approach for dynamic gesture recognition and sentence formation

The platform incorporates a structured learning framework divided into alphabet, word, and sentence levels. Users receive real-time feedback, accuracy evaluation, and performance tracking to improve learning efficiency. User authentication, progress tracking, and data management are handled using a secure PostgreSQL database with rowlevel security.

The developed system was tested for recognition accuracy, usability, responsiveness, and efficiency. The results demonstrated efficient real-time gesture recognition with low latency and strong performance on standard consumer devices. Since all processing is performed directly within the browser, the system also ensures user privacy by keeping webcam data local to the client device.

This project can be applied in educational institutions, assistive communication systems, accessibility-focused platforms, and inclusive learning environments. It also provides a strong foundation for future advancements in AI-driven gesture recognition systems and human-computer interaction technologies.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Project

Communication is one of the most important aspects of human interaction. For millions of Deaf and Hard-of-Hearing individuals, sign language acts as the primary medium of communication. However, many hearing individuals are unfamiliar with sign language, which creates communication barriers in schools, workplaces, hospitals, and social environments.

Traditional sign language recognition systems often depend on sensor-based gloves and expensive hardware devices. These systems are costly and difficult to use in real-world environments. Existing sign language learning applications mostly provide static learning content without real-time interaction or feedback.

The proposed project, SANKETA – Real-Time Sign Language Recognition and Learning System, aims to provide a browser-based intelligent platform that combines sign language recognition with interactive learning support.

1.2 Problem Statement

Communication barriers between hearing individuals and the Deaf and Hard-of-Hearing community continue to exist because of limited awareness and understanding of sign language. Existing systems either focus only on recognition or only on learning. Many systems require expensive hardware or lack real-time feedback mechanisms.

Therefore, there is a need for an intelligent, browser-based, and accessible system that can provide real-time sign language recognition along with structured learning support.

1.3 Objectives

The main objectives of this project are:

- To develop a browser-based sign language recognition system.
- To perform real-time hand gesture detection using computer vision techniques.
- To implement gesture classification using machine learning algorithms.
- To provide structured learning modules for alphabets, words, and sentences.

- To implement real-time feedback and performance tracking.
- To ensure privacy through client-side processing.

1.4 Scope of the Project

The scope of this project includes the development of a web-based platform capable of recognizing sign language gestures using a webcam. The system supports learning modules for users through interactive practice sessions and performance tracking.

The project includes gesture recognition, learning feedback, progress tracking, and continuous gesture interpretation. Future enhancements may include speech synthesis integration, multilingual support, and mobile application support.

1.5 Organization of the Report

The report is organized into different chapters for better understanding of the project.

- Chapter 1 explains the introduction, objectives, and scope of the project.
- Chapter 2 presents the literature review of existing systems and technologies.
- Chapter 3 describes the system analysis and proposed solution.
- Chapter 4 explains the system design and architecture.
- Chapter 5 explains implementation details and technologies used.
- Chapter 6 discusses results, testing, and performance analysis.
- Chapter 7 presents the conclusion and future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Existing Systems / Research

Sign language recognition has evolved significantly over the years with the development of computer vision and artificial intelligence technologies. Earlier systems mainly depended on hardware-based solutions such as sensor gloves and motion tracking devices to capture hand movements accurately. These systems provided good gesture recognition accuracy because sensors could directly measure finger and hand movements. However, they were expensive, uncomfortable to wear for long periods, and difficult to use in realworld situations.

To overcome these limitations, researchers later introduced camera-based sign language recognition systems. These systems used webcams and image processing techniques such as skin colour detection, contour extraction, and hand segmentation to identify hand gestures. Compared to sensor-based systems, camera-based methods were more affordable and easier to access because they did not require external hardware. However, their performance was highly affected by lighting conditions, background complexity, and camera quality.

With advancements in machine learning, many researchers started using algorithms such as Support Vector Machines (SVM) and k-Nearest Neighbour (KNN) for gesture classification. These algorithms improved gesture recognition by analysing extracted hand features more efficiently. KNN became popular for lightweight systems because of its simplicity and fast prediction capability.

Recent developments in deep learning introduced Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for sign language recognition. CNNs are effective in identifying spatial features from hand gesture images, while RNNs and LSTM models are useful for recognizing continuous and dynamic gestures. These approaches improved recognition accuracy significantly, especially for sentence-level gesture interpretation. However, deep learning models require large datasets, powerful hardware, and high computational resources, which make them difficult to deploy on low-end devices or browser-based platforms.

Modern frameworks such as Media Pipe Hands have further improved real-time gesture recognition by providing efficient hand landmark detection using standard webcams. Media Pipe can detect 21 hand landmarks in real time with high accuracy, making lightweight browser-based sign language systems possible without specialized hardware.

2.2 Comparison of Methods

Method/System	Working method	Advantages	Limitations
Sensor-based systems	Uses sensor gloves	High accuracy	Expensive hardware
Camera-based systems	Uses image processing	No external devices	Sensitive to lighting
Deep learning systems	Uses CNN and RNN models	High accuracy	High computational cost
Learning platforms	Provides tutorials	Easy learning access	No real-time feedback
Proposed SANKETA System	Combines recognition and learning	Real-time feedback, accessibility	Requires webcam support
Cloud-Based Recognition Systems	Sends video data to cloud servers for gesture processing	Powerful processing capabilities	Privacy concerns and higher latency

2.3 Limitations of Current Solutions

Existing systems face several limitations such as dependency on expensive hardware, lack of realtime feedback, high computational requirements, and limited accessibility.

Many learning platforms do not provide interactive evaluation or adaptive learning mechanisms. Existing recognition systems also lack privacy-focused browser-based execution.

CHAPTER 3:

SYSTEM ANALYSIS

3.1 Existing System

Existing sign language recognition systems mainly focus on either gesture recognition or sign language learning separately. Earlier systems depended heavily on hardware devices such as sensor gloves and motion tracking equipment to capture hand movements accurately. Although these systems provided good recognition accuracy, they were expensive, uncomfortable to use for long durations, and difficult to deploy on a large scale. The requirement of specialized hardware also reduced accessibility for normal users.

Later, camera-based sign language recognition systems were introduced using image processing and computer vision techniques. These systems used methods such as skin colour detection, contour extraction, and hand segmentation to recognize gestures through webcam input. While these approaches removed the dependency on hardware devices, they were highly affected by environmental conditions such as poor lighting, complex backgrounds, and camera quality. As a result, recognition accuracy often decreased in realworld environments.

Recent advancements introduced deep learning-based systems using Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for gesture recognition. These systems improved recognition accuracy significantly, especially for complex and dynamic gestures. However, deep learning models require large datasets, high computational power, and expensive processing resources. This makes them difficult to run efficiently on lowend devices or directly inside web browsers.

3.2 Proposed System

The proposed system, “SANKETA – Real-Time Sign Language Recognition and Learning System,” is a browser-based platform designed to provide both real-time sign language recognition and interactive learning support.

The system uses Media Pipe Hands to detect and track hand movements through a webcam. It extracts 21 hand landmarks and processes them using feature engineering techniques. A lightweight k-Nearest Neighbour (KNN) algorithm is used to classify gestures such as alphabets and words.

The system also includes a temporal processing mechanism to recognize continuous gestures and convert them into meaningful sentence outputs.

In addition to recognition, the platform provides structured learning modules for alphabets, words, and sentences. Users receive real-time feedback, accuracy scores, and progress tracking to improve learning.

All processing is performed directly inside the browser, which reduces latency and improves privacy by keeping webcam data on the user's device.

3.3 Advantages of Proposed System

- Browser-based system without specialized hardware.
- Real-time gesture recognition using a webcam.
- Lightweight and efficient machine learning model.
- Interactive learning with real-time feedback.
- Client-side processing improves privacy and reduces latency.
- Structured learning modules for better understanding.
- Progress tracking and performance evaluation.
- Scalable and accessible for different users.

Overall, the proposed system provides an efficient, accessible, and user-friendly solution for sign language recognition and learning.

CHAPTER 4:

SYSTEM DESIGN

4.1 Architecture Diagram

The SANKETA system follows a modular browser-based architecture designed for realtime sign language recognition and interactive learning. The system integrates computer vision, machine learning, and database management components to provide efficient gesture recognition and learning support.

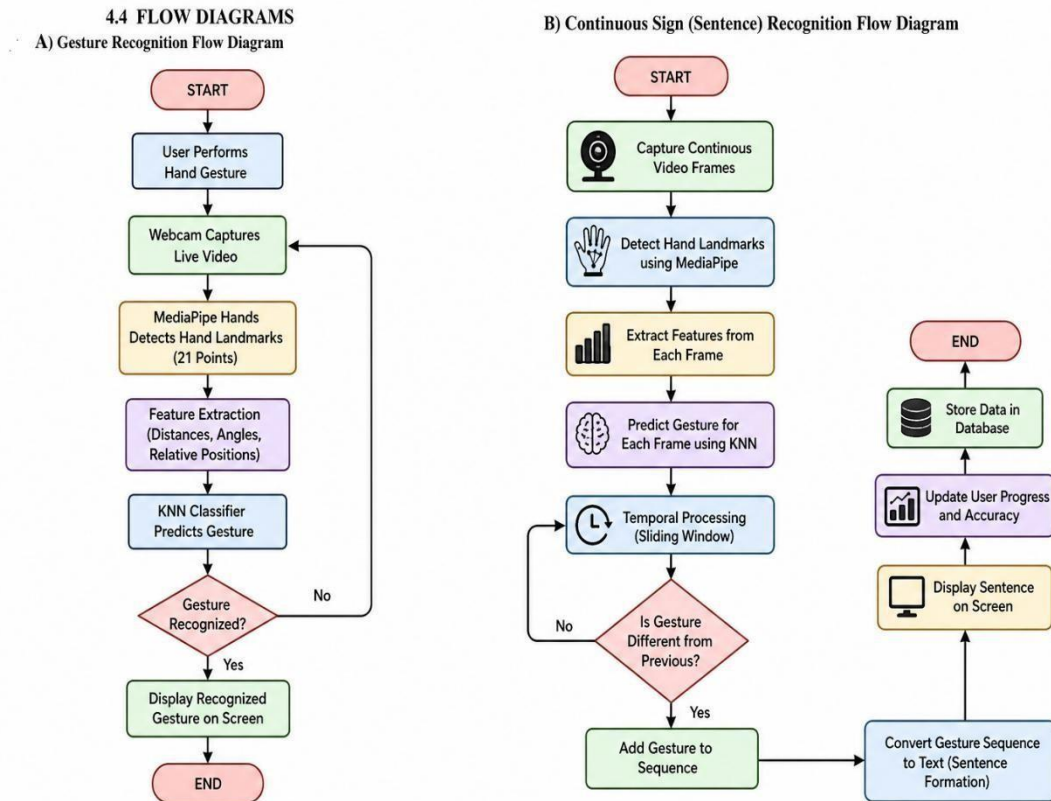


Fig 1A. Gesture Recognition Flow Diagram , 1B. Continuous Sign(Sentence) Recognition Flow Diagram

4.2 Data Flow Diagram (DFD)

The Data Flow Diagram explains how data moves between the different modules of the SANKETA system.

DFD Description

The user performs hand gestures in front of the webcam.

The webcam captures live video frames.

MediaPipe Hands detects hand landmarks from the frames.

The feature extraction module processes landmark data.

The KNN classifier predicts the gesture.

The predicted output is displayed to the user.

The learning module evaluates user accuracy.

User progress and scores are stored in the database.

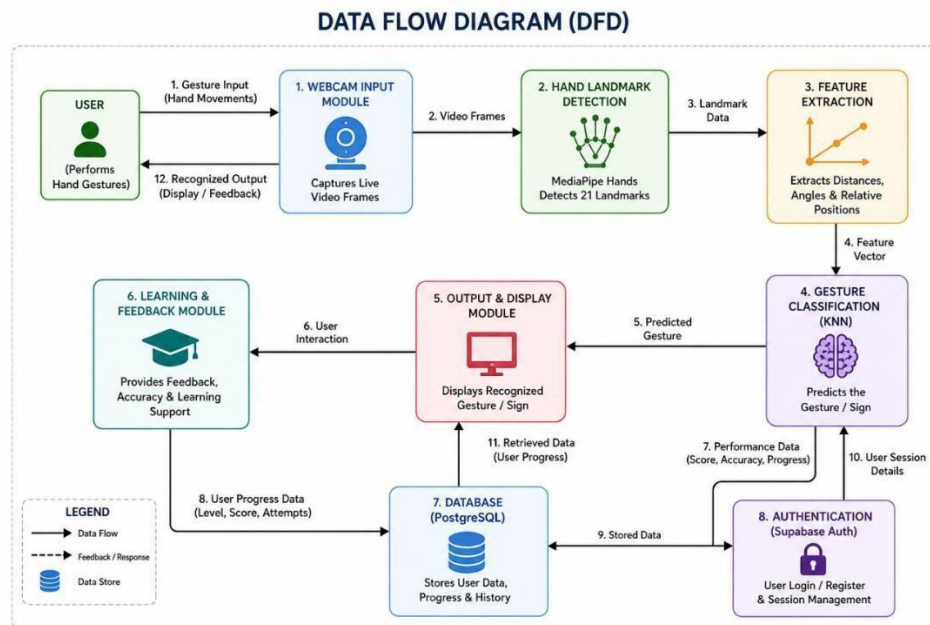


Fig 2. Data Flow Diagram

4.3 UML Diagrams

Use Case Diagram

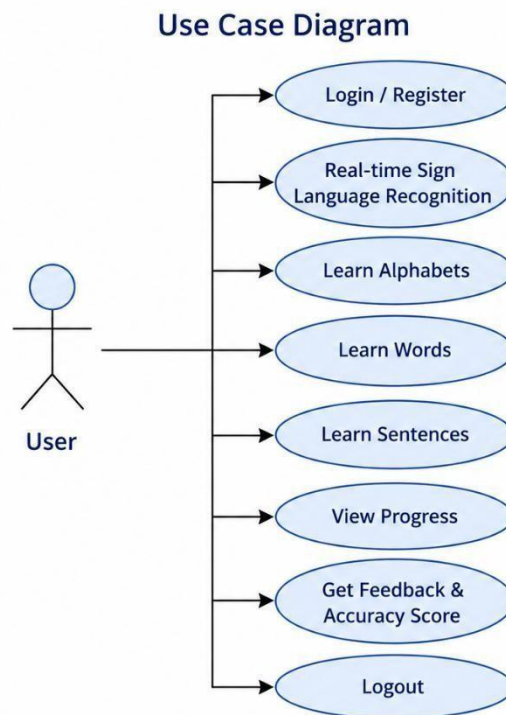


Fig 3. Use case diagram

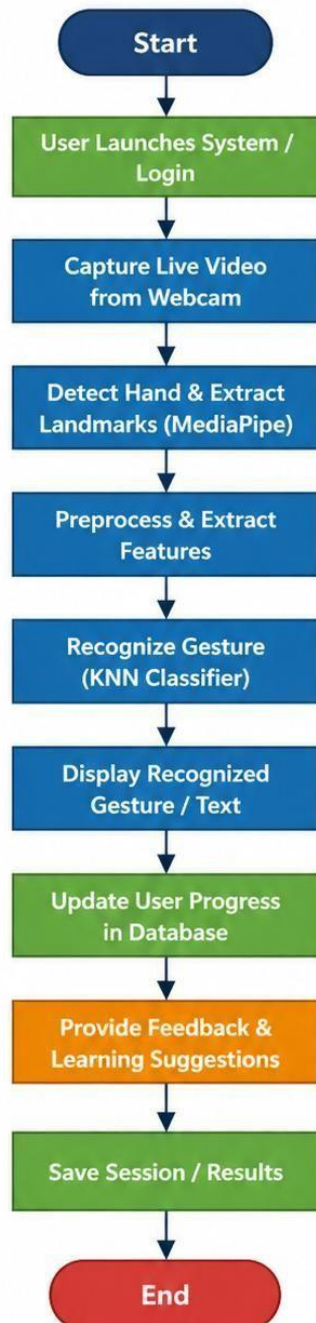
4.4 Algorithms / Flowcharts

The use case diagram represents the interaction between the user and the SANKETA system.

Main Functionalities

- User Registration
- User Login
- Real-Time Gesture Recognition
- Sign Language Practice
- Sentence Recognition
- View Progress
- Receive Feedback

Sign Language Recognition System Flowchart



Class Diagram

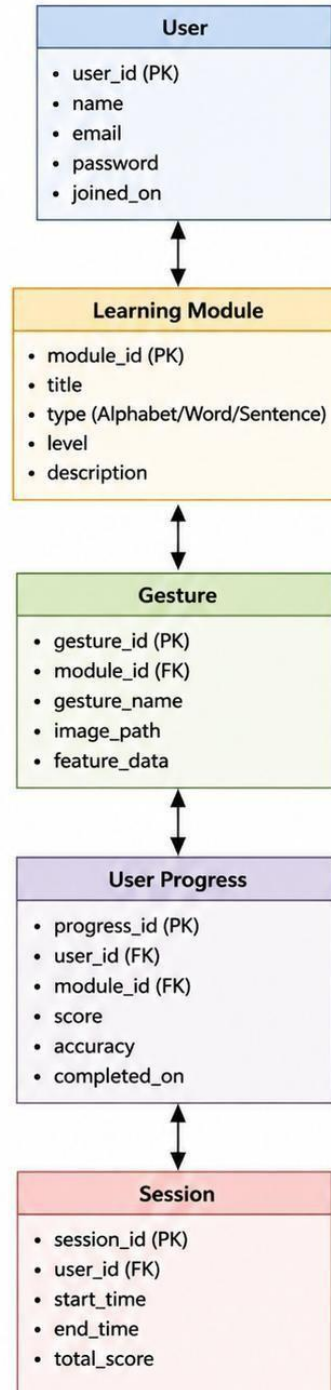


Fig 4. Sign language recognition System Flow chart and Class Diagram

CHAPTER 5

IMPLEMENTATION

5.1 Tools and Technologies Used

The proposed project, “SANKETA – Real-Time Sign Language Recognition and Learning System,” was developed using modern web technologies and machine learning frameworks to ensure real-time performance, accessibility, and efficient gesture recognition. The frontend of the system was developed using React.js with TypeScript, which provides a responsive and interactive user interface. Tailwind CSS was used for designing clean, modern, and user-friendly UI components.

The gesture recognition functionality was implemented using MediaPipe Hands, which detects and tracks 21 hand landmarks in real time using webcam input. OpenCV and JavaScript-based browser APIs were used for handling webcam video processing and frame capturing.

For gesture classification, the system uses the k-Nearest Neighbour (KNN) machine learning algorithm because of its lightweight nature and fast prediction capability. Feature engineering techniques such as distance calculation, angle detection, and relative landmark positioning were used to improve recognition accuracy.

The backend services were implemented using Supabase, which provides authentication, PostgreSQL database management, APIs, and secure data storage. Supabase also simplifies deployment and user management without requiring a separate backend server.

For routing and navigation between pages, React Router was used. State management and user progress handling were managed using Context API and React hooks. Browser webcam APIs were used to capture live hand gesture input directly from the user’s device.

5.2 Hardware / Software Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: 20 GB free disk space
- Webcam: Required for gesture recognition
- Internet Connection: Required

Software Requirements

- Operating System: Windows / Linux / macOS
- Web Browser: Chrome / Edge / Firefox
- Code Editor: Visual Studio Code
- Node.js and npm
- Supabase Account
- Python Environment (for model training if required)

5.3 Module Description

The project consists of the following major modules:

1. User Authentication Module

This module allows users to register, log in, and securely access the system using email credentials. Supabase Authentication is used for secure login and user management.

2. Hand Detection Module

This module captures live webcam input and detects hand landmarks using MediaPipe Hands technology.

3. Feature Extraction Module

This module processes hand landmark data and extracts important features such as distances, angles, and relative hand positions for gesture recognition.

4. Gesture Recognition Module

This module uses the KNN machine learning algorithm to classify hand gestures and predict the corresponding sign language output.

5. Continuous Gesture Processing Module

This module processes sequences of gestures using temporal analysis and sliding window techniques to generate meaningful sentence outputs.

6. Learning and Feedback Module

This module provides interactive learning support, evaluates user gestures, calculates accuracy scores, and provides feedback for improvement.

7. Progress Tracking Module

This module stores user learning progress, scores, and completed lessons in the PostgreSQL database.

5.4 Sample Code Snippets

Webcam Access for Gesture Capture

```
navigator.mediaDevices.getUserMedia({video:true})  
.then((stream)=>{video.srcObject=stream;  
});
```

MediaPipe Hand Landmark Detection

```
Consthands=newHands({  
  locateFile:(file)=>  
    `https://cdn.jsdelivr.net/npm/@mediapipe/hands/${file}` , });
```

Gesture Prediction using KNN

```
prediction=knn.predict(feature_vector) print(prediction)
```

Accuracy Calculation accuracy =

```
(correct_predictions / total_attempts) * 100
```

CHAPTER 6:

RESULTS AND DISCUSSION

6.1 Output Screenshots

The developed “SANKETA – Real-Time Sign Language Recognition and Learning System” was successfully implemented and tested through different modules. The important output screens obtained during execution are listed below:

1. Login / Registration Page

Allows users to create an account and securely log in to the system for accessing learning modules and gesture recognition features.

2. Real-Time Gesture Recognition Screen

Displays live webcam input and recognizes hand gestures in real time using MediaPipe Hands and the KNN classification model.

3. Hand Landmark Detection Screen

Shows the detected hand landmarks and finger tracking points extracted from the webcam feed.

4. Alphabet Learning Module

Allows users to practice sign language alphabets with real-time recognition and feedback.

5. Word Learning Module

Provides practice sessions for learning commonly used sign language words and gesture combinations.

6. Sentence Recognition Module

Recognizes continuous gesture sequences and converts them into meaningful sentence outputs.

7. Feedback and Accuracy Screen

Displays gesture prediction accuracy, correctness feedback, and learning performance analysis.

8. User Progress Dashboard

Shows user learning progress, completed lessons, accuracy scores, and performance statistics.

6.2 Performance Analysis

The system was evaluated based on recognition accuracy, responsiveness, usability, efficiency, and real-time performance. The results showed that the platform performs effectively under normal operating conditions.

- Login Response Time: Fast and secure authentication using Supabase authentication services.
- Gesture Recognition Speed: Real-time gesture prediction achieved with low latency during webcam processing.
- Hand Detection Accuracy: MediaPipe Hands accurately detected and tracked hand landmarks in real time.
- Gesture Classification Accuracy: The KNN model successfully recognized static gestures with good accuracy.
- Continuous Gesture Recognition: The temporal processing module effectively interpreted gesture sequences for sentence generation.
- User Interface Performance: Smooth and responsive interface across different devices and browsers.
- Scalability: The browser-based architecture supports multiple users without requiring specialized hardware.

The use of React.js, MediaPipe Hands, and lightweight machine learning models helped improve overall system performance while maintaining low computational requirements.

6.3 Test Cases and Results

Test case ID	Test Scenario	Expected Result	Actual Result	Status
TC01	User Registration	New user account created successfully	Account created	Pass
TC02	User Login	Valid user logs into dashboard	Login successful	Pass
TC03	Hand Landmark Detection	Hand landmarks detected correctly	Hand landmarks detected	Pass
TC04	Static Gesture Recognition	Correct gesture identified	Correct gesture predicted	Pass
TC05	Alphabet Learning Module	User receives correct feedback	Feedback displayed correctly	Pass
TC06	Word Recognition	Word gesture identified correctly	Word recognized successfully	Pass
TC07	Continuous Gesture Recognition	Sentence generated from gestures	Sentence generated correctly	Pass
TC08	Progress Tracking Module	User progress stored in database	Progress updated successfully	Pass

6.4 Discussion

The testing results confirm that the proposed “SANKETA – Real-Time Sign Language Recognition and Learning System” successfully integrates real-time sign language recognition and interactive learning into a single browser-based platform. All major modules functioned effectively during testing and provided reliable performance.

The hand detection module performed efficiently by accurately detecting and tracking hand landmarks using MediaPipe Hands technology. This enabled smooth real-time gesture recognition through webcam input without requiring external hardware devices.

The gesture classification module successfully recognized static gestures such as alphabets and words using the KNN machine learning algorithm. The system achieved good recognition accuracy while maintaining low latency and lightweight processing requirements.

The continuous gesture processing module effectively analyzed gesture sequences and converted them into meaningful sentence outputs using temporal analysis techniques. This improved the communication capabilities of the system beyond static gesture recognition.

The learning and feedback modules improved user engagement by providing structured learning levels, accuracy evaluation, and real-time correction feedback. The progress tracking system also helped users monitor their improvement over time.

The overall system was found to be user-friendly, responsive, scalable, and privacy-focused because all gesture processing was performed directly inside the browser. However, certain limitations such as dependency on lighting conditions, webcam quality, and limited gesture datasets were observed.

CHAPTER 7:

CONCLUSION AND FUTURE WORK

7.1 Summary of Work Done

The project “SANKETA – Real-Time Sign Language Recognition and Learning System” was developed to reduce communication barriers between hearing individuals and the Deaf and Hard-of-Hearing community through real-time gesture recognition and interactive learning support. The system was implemented as a browser-based platform that combines multiple features such as real-time hand gesture recognition, continuous gesture processing, structured learning modules, user authentication, progress tracking, and feedback mechanisms.

Modern technologies such as React.js, TypeScript, Tailwind CSS, MediaPipe Hands, and Supabase were used for frontend development, gesture processing, backend services, database management, and secure user authentication. The system uses MediaPipe Hands for detecting hand landmarks and a lightweight KNN machine learning algorithm for gesture classification. Temporal gesture processing techniques were also implemented to support sentence-level gesture recognition.

The platform was successfully tested and was found to be responsive, efficient, userfriendly, and capable of performing real-time sign language recognition with low latency on standard consumer devices.

7.2 Key Findings

The major findings observed from the project are:

- Real-time hand gesture recognition improves communication support for sign language users.
- Browser-based gesture recognition eliminates the need for specialized hardware devices.
- MediaPipe Hands provides accurate and efficient real-time hand landmark detection.
- Lightweight machine learning models such as KNN can achieve good recognition accuracy with low computational requirements.
- Structured learning modules improve user engagement and learning efficiency.
- Real-time feedback and accuracy evaluation help users improve sign language

skills effectively.

7.3 Limitations

Although the system performed successfully, certain limitations were identified:

- Recognition accuracy may reduce under poor lighting conditions.
- Webcam quality can affect hand landmark detection performance.
- The current system supports a limited set of gestures and datasets.
- Dynamic gesture recognition can be improved for more complex sentence interpretation.
- Continuous gesture recognition may sometimes produce incorrect predictions during fast hand movements.
- The current implementation is browser-based and does not include a dedicated mobile application.

7.4 Future Enhancements

The project can be improved further with the following enhancements:

- Support for multiple sign languages and regional sign variations. •
Development of dedicated Android and iOS mobile applications.
- Speech synthesis integration for converting recognized gestures into voice output.
- Offline support for gesture recognition without internet dependency.
- Expansion of gesture datasets for better recognition performance.
- AI-based personalized learning recommendations for users.

7.5 Conclusion

Overall, the project demonstrates how computer vision, machine learning, and modern web technologies can be effectively combined to create an intelligent, accessible, and scalable sign language recognition and learning system. The developed solution can play an important role in improving communication accessibility and promoting inclusivity for the Deaf and Hard-of-Hearing community.

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APPENDICES

APPENDIX A: SAMPLE CODE

A.1 User Login Authentication Code

```
const handleLogin = async () => {  
  const { error } = await supabase.auth.signInWithPassword({  
    email: email, password: password  
  });  
  
  if (!error) {  
    alert("Login Successful");  
  }  
};
```

A.2 Webcam Access for Gesture Recognition

```
navigator.mediaDevices.getUserMedia({ video: true })  
  .then((stream) => { video.srcObject = stream;  
  });
```

A.3 MediaPipe Hand Landmark Detection

```
const hands = new Hands({  
  locateFile: (file) => `https://cdn.jsdelivr.net/npm/@mediapipe/hands/${file}`,  
});  
hands.onResults(onResults);
```

A.4 Gesture Prediction using KNN

```
prediction = knn.predict(feature_vector) if prediction:  
  print("Gesture Recognized")
```

A.5 Accuracy Calculation

```
accuracy      =      (correct_predictions /      total_attempts)      *      100  
print("Accuracy:", accuracy)
```

APPENDIX B: ADDITIONAL DIAGRAMS

B.1 System Architecture Diagram

User → Webcam Input → MediaPipe Hands → Feature Extraction → KNN Classifier → Output Display → PostgreSQL Database

B.2 Gesture Recognition Flow Diagram

Start → Capture Webcam Input → Detect Hand Landmarks → Extract Features → Predict Gesture → Display Output → End

B.3 Continuous Gesture Processing Flow

Capture Gesture Sequence → Analyze Frames → Remove Duplicate Predictions → Detect Gesture Transition → Generate Sentence Output

B.4 Module Interaction Diagram

Login Module ↔ Gesture Recognition Module ↔ Learning Module ↔ Feedback Module
↔ Progress Tracking Module ↔ Database

APPENDIX C: DATA SETS

The project uses hand landmark datasets and real-time webcam-generated gesture data for training and testing the gesture recognition model.

C.1 User Profile Data

User ID	Name	Email
---------	------	-------

U001	User 1	user1@gmail.com
U002	User 2	user2@gmail.com

C.2 Gesture Training Data

Gesture ID	Gesture Name	Category
G001	A	Alphabet
G002	Hello	Word
G003	Thank You	Sentence

C.3 Hand Landmark Data

Landmark ID	X Coordinate	Y Coordinate	Z Coordinate
L001	0.45	0.62	-0.03
L002	0.51	0.58	-0.02

C.4 User Progress Data

User ID	Completed Level	Accuracy Score
U001	Alphabet Level	92%
U002	Word Level	88%

C.5 Gesture Prediction Results

Input Gesture	Predicted Output	Status
Hand Sign A	A	Correct
Hand Sign Hello	Hello	Correct
Sentence Gesture	Welcome	Correct

These appendices provide additional technical details, sample code snippets, flow diagrams, and datasets used for the successful implementation of the “SANKETA – RealTime Sign Language Recognition and Learning System.